

PAULA FUDOLIG

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RESEARCH INTERESTS

Multi-wavelength studies of black hole – galaxy co-evolution.

EDUCATION

George Mason University <i>Doctor of Philosophy in Physics</i>	Aug. 2023 – Present <i>Fairfax, VA</i>
George Mason University <i>Master of Science in Applied and Engineering Physics</i>	Aug. 2023 – May 2026 <i>Fairfax, VA</i>
Boston University <i>Bachelor of Arts in Astronomy and Physics</i>	Aug. 2019 – May 2023 <i>Boston, MA</i>

PROJECT EXPERIENCE

Radio Emission in Radio-Quiet Quasars <i>Graduate research, in collaboration with the National Radio Astronomy Observatory</i> - VLA data analysis of optically-selected radio-quiet quasars to search for sub-galactic emission. - VLBA data reduction in AIPS, imaging, and analysis of optically-selected radio-quiet quasars to quantify AGN contributions to the observed radio emission.	May 2025 – Present <i>Fairfax, VA</i>
Varstrometry for Dual AGN using Radio Interferometry <i>Graduate research, in collaboration with the US Naval Research Lab</i> VLBA data reduction in CASA, imaging, and analysis of radio-loud quasar observations at 2 GHz and 8 GHz to investigate varstrometry+radio interferometry as a viable systematic detection method for multi-AGN.	June 2023 – Present <i>Fairfax, VA</i>
Large VLBA BEAM-ME <i>Boston University Institute of Astrophysical Research</i> VLBA data reduction in Difmap of monthly blazar observations to trace the motion of shocks propagating from the jet base and measure their statistical correlation to simultaneous gamma flares seen in Fermi-LAT data.	Jan. 2020 – May 2023 <i>Boston, MA</i>
Patatrack Pixel Tracking <i>European Council for Nuclear Research (CERN)</i> Tested Patatrack's 4 portability libraries of track reconstruction algorithms across 3 GPU devices on a varying amount of threads to evaluate performance, using CUDA in C++ and Python.	Jan. 2022 – Aug. 2022 <i>Geneva, Switzerland</i>
SOFIA 3-Micron Dust Emission Studies <i>NASA Goddard Space Flight Center</i> Fitted 3-5 micron wavelength emission spectra in Python from polycyclic aromatic hydrocarbons surrounding a sample of planetary nebulae to study the relationship between hydrocarbon features and planetary nebulae ages.	June 2020 – Aug. 2020 <i>Greenbelt, MD</i>

PUBLICATIONS

Schwartzman, E., **Fudolig, P.**, et al. Varstrometry for Dual AGN using Radio Interferometry: VaDAR with the VLBA, 2025, ApJ, 987, 200

PROPOSALS

Probing Variability in Radio-Quiet (Compared to Radio-Loud) QSOs Co-investigator, awarded 77.5 hours at 6 and 10 GHz on the VLA in C-configuration.	Semester 2026B
Tracing the Origin of AGN Emission in Variable Radio-Quiet Quasars Principal investigator, awarded 24.5 hours at 1, 6, and 8 GHz on the VLBA.	Semester 2026A
VLBA Characterization of Astrometrically-Variable Quasars Principal investigator, awarded 24 hours at 1, 6, and 15 GHz on the VLBA.	Semester 2024B

GRANTS

Undergraduate Research Opportunities Program

June 2022 – Aug. 2022

- > \$4000 in funding from Boston University to extend internship at CERN.

PRESENTATIONS & POSTERS

(* = invited; † = poster, otherwise presentation)

Quiet Chaos: Radio Emission in Variable Radio-Quiet Quasars* <i>National Radio Astronomy Observatory</i>	Jul. 22, 2026 <i>Socorro, NM</i>
A Tale of Two Variabilities: VaDAR and the Brightening J0822+4553* <i>National Radio Astronomy Observatory</i>	Jan. 21, 2026 <i>Socorro, NM</i>
Tracing the Origins of Radio Emission in Radio-Quiet Quasars <i>DC Astrophysics Graduate Student Conference</i>	Aug. 14-15, 2025 <i>Washington, D.C.</i>
A VLBA View of VaDAR <i>245th American Astronomical Society Meeting</i>	Jan. 12-16, 2025 <i>National Harbor, MD</i>
A VLBA View of VaDAR <i>11th Annual Mid-Atlantic Radio-Loud AGN Meeting</i>	November 8, 2024 <i>Baltimore, MD</i>
A VLBA View of VaDAR <i>DC Astrophysics Graduate Student Conference</i>	Aug. 15-16, 2024 <i>Washington, D.C.</i>
VaDAR: Varstrometry for Dual AGN using Radio Interferometry <i>10th Annual Mid-Atlantic Radio-Loud AGN Meeting</i>	October 12, 2023 <i>Fairfax, VA</i>
Pixel-Tracking in CERN's Compact Muon Solenoid† <i>Boston University Undergraduate Research Symposium</i>	Oct. 21, 2022 <i>Boston, MA</i>
Comparing Fits of PAH Emission <i>NASA Goddard Summer Intern Symposium</i>	August 4, 2020 <i>Greenbelt, MD</i>

SERVICE

Student Organization President <i>George Mason University SPECTRUM</i>	March 2024 – Present <i>Fairfax, VA</i>
• Co-led the 'Humans of Physics' project, involving personal interviews of faculty, staff, and students within the physics department in an effort to highlight the humanity of scientists. • Organized multiple 'Physics Cafe' events to foster community within the physics department. • Organized a series of lunch workshops promoting the professional development of graduate and undergraduate students within the College of Science. • Created and implemented a mentorship initiative for prospective graduate students of the physics PhD program. Resulted in a 100% retention rate upon its first implementation in Spring 2024.	
Dean's Advisor <i>George Mason University College of Science Dean's Student Advisory Board</i>	August 2025 – Present <i>Fairfax, VA</i>
• Shared insights and feedback on the student experience, representing graduate student needs and interests. • Collaborated with the Dean's office on college-level initiatives and improvements. • Fostered a stronger sense of community across programs and disciplines within the College of Science.	
Physics Department Representative <i>George Mason University Graduate and Professional Student Association</i>	August 2024 – May 2025 <i>Fairfax, VA</i>
• Advocated for fellow Physics graduate students on topics including graduate tuition and fees, fellowship opportunities, student life, and campus events. • Collaborated with other graduate student representatives from all departments on University-level initiatives. • Communicated important information on departmental changes and solicited feedback for upcoming votes.	
Peer Mentor <i>Boston University PeeRs for Incoming Student Mentorship Program</i>	Sep. 2021 – May 2023 <i>Boston, MA</i>

- Guided first-year students by providing departmental and campus resources, creating four-year plans, sharing personal experiences, and providing support throughout the academic year.

Student Organization Founder/President

Aug. 2020 – May 2023

Boston University chapter of the Society of Women in Space Exploration

Boston, MA

- As Founder: authored organization's constitution and bylaws, assembled an executive board, recruited a faculty advisor, and advertised on campus to build membership.
- As President: led biweekly general member meetings, involving devising group activities, planning on-campus events, and networking in the Boston area for collaborations. Also led executive board meetings, delegated tasks, and oversaw management of responsibilities.

RELATED EXPERIENCE

Graduate Teaching Assistant

Aug. 2023 – Present

- Led undergraduate physics and astronomy labs at George Mason University, typically 2-3 sections of 24-27 students per semester.
- Lab course taught (number of semesters): University Physics I (2); University Physics II (6); Stars, Galaxies, and the Universe (1)

Astronomy on Tap Speaker

November 18, 2025

- Presented "Hunting for Dual AGN with VaDAR" to the general public in Washington, DC.

Observation Time on Perkins Telescope

March 8-10, 2023

- Three nights operating Boston University's Perkins Telescope at Anderson Mesa Observatory in Flagstaff, AZ.

SCIENTIFIC & TECHNICAL SKILLS

Programming: Python, C++, CUDA, ROOT, Git, LaTeX, Ubuntu, Jupyter Notebook, Mathematica, Bash/Shell

Astronomical: NRAO CASA, NRAO AIPS, Caltech Difmap, PyBDSF, TopCat, SAOImage DS9, Maxim DL

World Languages: English (native), Spanish (advanced), French (intermediate), Tagalog (basic)